# Back to 2D Heat Transfer

## The Strong Form:

$div(\mathbf{K} \cdot \text{grad}(T)) + Q = 0$

with boundary conditions:

$T = T_A \text{ on } \partial\Omega_A, q = q^* \text{ on } \partial\Omega_B$

where $\mathbf{K}$ is the thermal conductivity tensor, $Q$ is the heat source term, and $q^*$ is a heat flux boundary condition.

## The Weak Form:

$$\int_\Omega \text{grad}(V) \cdot \mathbf{K} \cdot \text{grad}(T)\, d\Omega = \int_\Omega QV\, d\Omega - \int_{\partial\Omega_B} V q^*\, da$$

## Test and Trial Functions:

We choose to take both test and trial functions from the same function space, which is spanned by the shape functions $\phi_i$ defined via the master element coordinates $(\zeta_1, \zeta_2)$ and master element shape functions $\hat{\phi}$.

$$T = \sum_{j=1}^{N} a_j \phi_j$$

$$V = \sum_{i=1}^{N} b_i \phi_i$$

Also, we replace the integral over sums with sum over integrals, leading to:

$$\sum_{i=1}^{N} b_i \left[ \sum_{j=1}^{N} \left( \int_\Omega \text{grad}\phi_i \cdot \mathbf{K} \cdot \text{grad}\phi_j\, d\Omega \right) a_j \right] - \sum_{i=1}^{N} b_i \left( \int_\Omega \phi_i Q d\Omega - \int_{\partial\Omega_B} \phi_i \right.$$

# 2D Heat Transfer contd.

The derived equation:

$$\sum_{i=1}^{N} b_i \left[ \sum_{j=1}^{N} \left( \int_{\Omega} \text{grad}\phi_i \cdot \mathbf{K} \cdot \text{grad}\phi_j \, d\Omega \right) a_j \right] - \sum_{i=1}^{N} b_i \left( \int_{\Omega} \phi_i Q d\Omega - \int_{\partial\Omega_B} \phi_i \right.$$

represent a summation of the **element** integrals over **all nodes**

For elemnt $e$ whose nodes are $I, J$ :

$$K_{IJ}^{e} = \int_{\Omega^e} \text{grad}\phi_I \cdot \mathbf{K} \cdot \text{grad}\phi_J \, d\Omega$$

Assuming isotropic material with constant thermal conductivity $\mathbf{K} = k\mathbf{I}$, we can simplify the integral:

$$K_{IJ}^{e} = k \int_{\Omega^e} \text{grad}\phi_I \cdot \text{grad}\phi_J \, d\Omega$$

This leads to the element stiffness matrix:

$$[\mathbf{k}^e] = k \int_{\Omega^e} \mathbf{B}^T \mathbf{B} \, d\Omega$$

where $\mathbf{B}$ is the B matrix containing the derivatives of the shape functions $\phi_i$ with respect to global coordinates (x,y).

$$[\mathbf{B}\phi^e] = \begin{bmatrix} \frac{\partial \phi_1}{\partial x_1} & \frac{\partial \phi_1}{\partial x_2} \\ \frac{\partial \phi_2}{\partial x_1} & \frac{\partial \phi_2}{\partial x_2} \\ \frac{\partial \phi_3}{\partial x_1} & \frac{\partial \phi_3}{\partial x_2} \\ \frac{\partial \phi_4}{\partial x_1} & \frac{\partial \phi_4}{\partial x_2} \end{bmatrix}$$

Such that:

$$K^e = \int_{\Omega^e} k[B\phi^e][B\phi^e]^T \, d\Omega$$

# 2D Heat Transfer contd.

As we saw in the previous lecture, using isporametric mapping, we can express the shape functions in terms of master element coordinates $(\zeta_1, \zeta_2)$ which will be used for interpolating the space as well.

$$\mathbf{x}(\zeta_1, \zeta_2) = \sum_{I=1}^{n_{\text{nodes}}} x_I \hat{\phi}_I(\zeta_1, \zeta_2)$$

$$\frac{\partial \phi_I}{\partial x_j} = \sum_{k=1}^{dim} \frac{\partial \hat{\phi}_I}{\partial \zeta_k} \frac{\partial \zeta_k}{\partial x_j}$$

The term $[\mathbf{B}\hat{\phi}^e]_{Ij}$ can be expressed as:

$$\frac{\partial \hat{\phi}_I}{\partial x_j} = \sum_{k=1}^{dim} \frac{\partial \phi_I}{\partial x_j} \frac{\partial x_j}{\partial \zeta_k}$$

or:

$$\mathbf{B}\hat{\phi}^e = \mathbf{B}\phi^e\mathbf{F}$$

$$\mathbf{B}\phi^e = \mathbf{B}\hat{\phi}^e\mathbf{F}^{-1}$$

With:

$$F_{ij} = \frac{\partial x_i}{\partial \zeta_j} = \sum_{I=1}^{nodes} x_I^j \frac{\partial \hat{\phi}_I(\zeta)}{\partial \zeta_j}$$

The determinant of $\mathbf{F}$ $(J)$is used to transform between the area in the master element and the area in the global coordinates:

$$d\Omega = Jd\hat{\Omega}$$

where $J = \det(\mathbf{F})$ and $d\hat{\Omega}$is the area in the master element coordinates.

We can now write all the terms in the weak form at the element level:

$$\mathbf{K}^e = \int_{\Omega^e} \mathbf{K}(\mathbf{x}(\zeta))(\mathbf{B}\hat{\phi}^e)(\mathbf{F}^T\mathbf{F})^{-1}(\mathbf{B}\hat{\phi}^e)^T J d\hat{\Omega}$$

$$\mathbf{f}^e = \int_{\Omega^e} \hat{\phi}^e(\zeta) f(\mathbf{x}(\zeta)) J d\hat{\Omega}$$

## 2D Heat Transfer Contd.

The only thing we have left is to define the flux boundary condition $\int_{\partial\Omega_B} \phi_i q^* da$ in terms of the master element coordinates.

We start with relating $da$ (global) and $dA$ (local):

$$d\zeta = \frac{d\zeta}{dA} dA = \mathbf{M}\, dA$$

$$d\mathbf{x} = \frac{d\mathbf{x}}{da} da = \mathbf{m}\, da$$

$$dx_i = \sum_{j=1}^{dim} \frac{\partial x_i}{\partial \zeta_j} d\zeta_j = \sum_{j=1}^{dim} F_{ij} d\zeta_j \Rightarrow d\mathbf{x} = \mathbf{F} d\zeta$$

Next :

$$d\zeta \cdot d\zeta = dA^2 \quad ; \quad d\mathbf{x} \cdot d\mathbf{x} = da^2$$

$$da^2 = \mathbf{F} d\zeta \cdot \mathbf{F} d\zeta = dA^2 (\mathbf{FM}) \cdot (\mathbf{FM})$$

And so,

$$\frac{da}{dA} = \sqrt{(\mathbf{FM})^T (\mathbf{FM})}$$

We can now write the boundary integral in terms of the master element coordinates:

$$Q_i^s = -\int_{\partial\Omega_B} \phi_i q^* da = -\int_{\partial\hat{\Omega}_B} \hat{\phi_i}(\zeta) q^*(\zeta(\mathbf{x})) \sqrt{(\mathbf{FM})^T (\mathbf{FM})} dA$$

Th flux B.C will be applied on some line of our 2D element, which means that either $\zeta_1$ or $\zeta_2$ will be constant and equal to $\pm 1$.

### 2D Finite Element Assembly

Algorithm 1: 2D Finite Element Assembly

Input: Number of elements $N_e$, Gauss points $n_g$

Output: Global stiffness matrix $\mathbf{K}$ and force vector $\mathbf{F}$

for $\text{elem} = 1$ to $N_e$ do

Set $\mathbf{K}^e = \mathbf{0}, \mathbf{f}^e = \mathbf{0}$; Form $\mathbf{X}^e$

for $m = 1$ to $n_g$ do (Gauss points)

for $n = 1$ to $n_g$ do

$\boldsymbol{\zeta} = [\zeta_1, \zeta_2] = [\text{Points}[m], \text{Points}[n]]$

Evaluate $\mathbf{x}(\boldsymbol{\zeta}) = (\mathbf{X}^e)^T \hat{\boldsymbol{\phi}}(\boldsymbol{\zeta}), \mathbf{F}, \det(\mathbf{F})$

$$\mathbf{K}^e \leftarrow \mathbf{K}^e + w_m w_n K(\mathbf{x}(\boldsymbol{\zeta}))\mathbf{D}\hat{\boldsymbol{\phi}}(\mathbf{F}^T\mathbf{F})^{-1}(\mathbf{D}\hat{\boldsymbol{\phi}})^T \det(\mathbf{F})$$

$$\mathbf{f}^e \leftarrow \mathbf{f}^e + w_m w_n f(\mathbf{x}(\boldsymbol{\zeta}))\hat{\boldsymbol{\phi}}(\boldsymbol{\zeta}) \det(\mathbf{F})$$

end for

end for

Assemble into global matrix $[\mathbf{K}]N \times N$ and vector $[\mathbf{F}]N \times 1$

end for

## Continuum Mechanics - A Reminder

In continuum mechanics, we deal with continuous media and their deformation under external forces.

- **Displacement $\mathbf{u}$**: The change in position of a point in the body, defined as $\mathbf{u} = \mathbf{x} - \mathbf{x}_0$, where $\mathbf{x}_0$ is the original position often written using capital $\mathbf{X}$.
- **Configuration**
- $\mathbf{X}$: The original position of a point in the body before deformation. with $E_i$ being the coordinate system in the reference configuration. $\mathbf{X} = X_i E_i$
- $\mathbf{x}$: The current position of the same point after deformation, with $e_i$ being the coordinate system in the current configuration $\mathbf{x} = x_i e_i$.
- **Deformation Gradient $\mathbf{F}$**: A tensor that describes the local deformation of the material. It relates the current configuration to the reference configuration:

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} = \begin{bmatrix} \frac{\partial x_1}{\partial X_1} & \frac{\partial x_1}{\partial X_2} & \cdots \\ \frac{\partial x_2}{\partial X_1} & \frac{\partial x_2}{\partial X_2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- **Strain $\boldsymbol{\varepsilon}$**: A measure of deformation defined as the symmetric part of the deformation gradient:

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\mathbf{F}^T\mathbf{F} - \mathbf{I})$$

where $\mathbf{I}$ is the identity tensor. Under the assumption of small deformations, the strain can be approximated as:

$$\boldsymbol{\varepsilon} \approx \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T) = \nabla^s \mathbf{u}$$

where $\nabla \mathbf{u}$ is the gradient of the displacement field and $\nabla^s \mathbf{u}$ is the symmetric gradient.

- **Stress $\boldsymbol{\sigma}$**: A measure of internal forces within the material, defined using a constitutive relation that relates stress to strain. For linear elasticity, this is given by:
  $$\boldsymbol{\sigma} = \mathbf{C} : \boldsymbol{\varepsilon}$$
  where $\mathbf{C}$ is the elasticity tensor (material property matrix) and $\boldsymbol{\sigma}$ is Caucy stress tensor.
- **tractions $\mathbf{t}$**: The force per unit area acting on a surface, related to stress by:
  $$\mathbf{t} = \boldsymbol{\sigma} \cdot \mathbf{n}$$
  where $\mathbf{n}$ is the outward normal to the surface.

## Continuum Mechanics contd.

- **Mass Conservation**: The principle that mass is conserved in a closed system. In continuum mechanics, this is expressed as:
  $$\int_\Omega \rho \, dv = \text{constant} = \int_\Omega \rho_0 \, dV \Rightarrow \rho J = \rho_0$$
  where $\rho$ is the current density, $\rho_0$ is the reference density, and $J = \det(\mathbf{F})$ is the Jacobian determinant of the deformation gradient.
- **Balnace of linear momentum**
  $$\int_\Omega \rho \frac{\partial^2 \mathbf{x}}{\partial t^2} \, dv = \int_\Omega \mathbf{f} \, dv + \int_{\partial\Omega} \mathbf{t} \, da$$
  This leads to the equation of motion:
  $$\operatorname{div}(\boldsymbol{\sigma}) + \mathbf{f} = \rho \ddot{\mathbf{u}}$$
- **Balance of angular momentum**:
  $$\frac{d}{dt} \int_\Omega \mathbf{x} \times (\rho v) \, dv = \int_\Omega \mathbf{x} \times \mathbf{f} \, dv + \int_{\partial\Omega} \mathbf{x} \times \mathbf{t} \, da$$

# Continuum Mechanics contd.

## Balnace of linear momentum:

Assuming a quasi-static case:

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = \rho \ddot{\mathbf{u}} = 0$$

with

$$\nabla \cdot \boldsymbol{\sigma} = \left( \frac{\partial \sigma_{ij}}{\partial x_j} \right) \mathbf{e}_i$$

where $\sigma_{ij}$ are the components of the stress tensor.

## Weak Form

We multiply by the test function and make use of

$$\nabla \cdot (\boldsymbol{\sigma} \cdot \mathbf{v}) = (\nabla \cdot \boldsymbol{\sigma}) \cdot \mathbf{v} + \boldsymbol{\sigma} : \nabla(\mathbf{v})$$

with

$$\boldsymbol{\sigma} : \nabla(\mathbf{v}) = \sigma_{ij} \frac{\partial v_j}{\partial x_i}$$

Speaker notes